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Thermodynamic assessments of the Sc–M (M: Ag, B and Th) systems

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Abstract

The phase diagrams and thermodynamic properties in the Sc–M (M: Ag, B and Th) binary systems have been assessed by using the CALPHAD (*Cal*culation of *Pha*se *D*iagrams) method on the basis of the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc, and hcp phases were described by the subregular solution model with the Redlich–Kister equation, and those of the intermetallic compounds (Ag_4Sc, Ag_2Sc, AgSc, ScB_2 and ScB_{12}) in these three binary systems were described by the sublattice model. The thermodynamic parameters of the Sc–M (M: Ag, B and Th) binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.
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Keywords: Phase diagrams; Transition metal alloys and compounds; Thermodynamic modeling

1. Introduction

Sc-based alloys have been used as aero-structures materials [1], and Sc addition on wrought alloys has positive effects on weldability and welding properties [2]. The phase equilibria in the Sc-based alloy systems are of importance for design of new Sc-based alloys. The CALPHAD method, which is a powerful tool to decrease cost and time during development of materials [3], effectively provides a clear guideline for material design. In the present paper, the thermodynamic descriptions for the phase diagrams in the Sc–M (M: Ag, B and Th) systems were carried out by combining thermodynamic models with the experimental data in the framework of the CALPHAD method.

2. Experimental information

2.1. The Sc–Ag system

The Sc–Ag system is consisted of three solution phases (fcc (Ag), αSc and βSc phases) and three intermetallic compounds (AgSc, Ag_2Sc and Ag_4Sc phases). Gschneidner et al. [4] studied the phase diagram in the Ag-rich region, and found the Ag_4Sc compound. Mcmasters et al. [5] reported that the Ag_4Sc compound has tetragonal structure ($MoNi_4$ type) by X-ray powder diffraction methods. Gschneidner et al. [4] also determined the temperature of the eutectic reaction (L ↔ (Ag) + Ag_4Sc) to be 926 ± 3 °C and the terminal solid solubility in fcc (Ag) solution phase to be 10.4 at.% Sc. Reule et al. [6] reported the lattice parameters and structure data of the AgSc, Ag_2Sc and Ag_4Sc intermetallic compounds by X-ray diffraction.

Later, based on the thermal analysis, X-ray diffraction and microscopic methods, the Sc–Ag system was investigated by Stapf et al. [7], who reported the temperature of the peritectic reaction (L + Ag_2Sc ↔ Ag_4Sc) to be 920 °C and the maximum solid solubility of Sc in fcc (Ag) solution phase to be 3–3.5 at.% at 910 °C. Stapf et al. [7] also determined three eutectic reactions: L ↔ (Ag) + Ag_4Sc at 10 at.% Sc and 910 °C, L ↔ Ag_2Sc + AgSc at 40 at.% Sc and 1130 °C, and L ↔ AgSc + αSc at 80 at.% Sc and 920 °C. The allotropic transformation of αSc ↔ βSc at 1337 °C was not indicated in Ref. [7].

The major difference between the reports of Gschneidner et al. [4] and Stapf et al. [7] was the maximum solid solubility of Sc in the fcc (Ag) solution phase (10.4 or ∼3.5 at.% Sc) and the eutectic temperature (L ↔ (Ag) + Ag_4Sc at 926 or 910 °C). The solid solubility and the eutectic temperature reported by Gschneidner et al. [4] were accepted in the Sc–Ag phase diagram compiled by Gschneidner and Calderwood [8] with some modifications from the experimental data reported by Stapf et al. [7]. The phase diagram of the Sc–Ag system reviewed by Gschneidner and Calderwood [8] is shown in Fig. 1.

In addition, Fitzner et al. [9] measured the standard enthalpies of formation of the AgSc and Ag_2Sc phases by direct high-temperature reaction calorimetry, and also determined the enthalpies of mixing of liquid phase at 1200 °C with reference states of pure elements of Ag (liquid) and Sc (liquid).

2.2. The Sc–B system

The Sc–B system is consisted of three solution phases (αSc, βSc and βB phases) and two intermetallic compounds (ScB_2 and ScB_{12} phases). The phase

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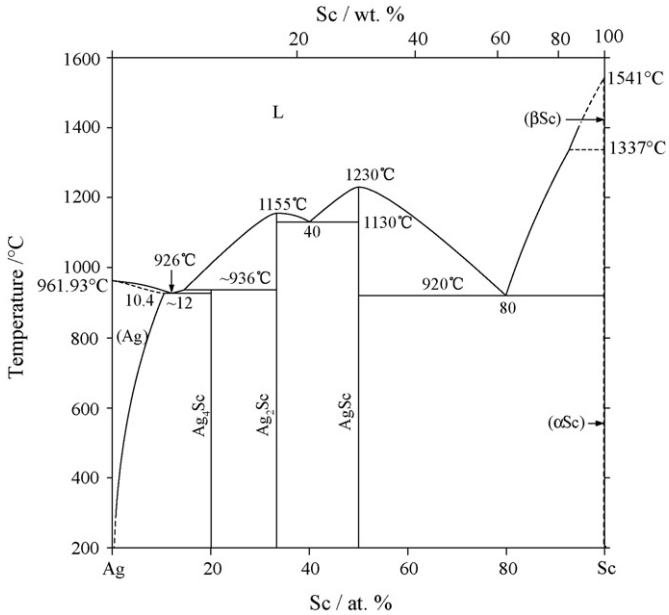

Fig. 1. The phase diagram of the Sc–Ag system reviewed by Gschneidner and Calderwood [8].

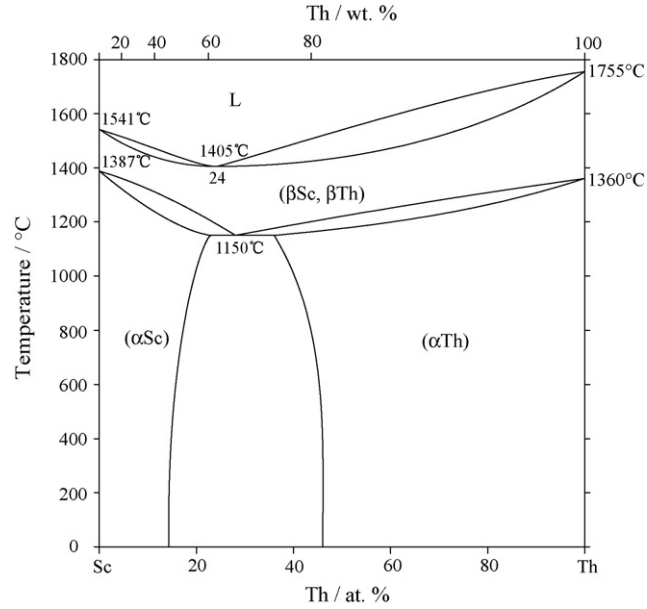

Fig. 3. The phase diagram of the Sc–Th system reviewed by Okamoto [17].

diagram is based on the experimental data [10–12], and liquidus line was calculated from the Gibbs energy functions optimized by Spear and Liao [13]. Peshev et al. [10] reported that the measurable solubility of Sc in rhombohedral βB is about 1 at.% Sc, while Callmer [11] reported a solubility of 3.5 at.% Sc in rhombohedral βB. Spear and Liao [13] expected the solubility of B in αSc and βSc to be small (<1 at.%). The phase diagram of the Sc–B system reviewed by Spear and Liao [13] is shown in Fig. 2.

In addition, Topor and Kleppa [14] measured the standard enthalpies of formation of the ScB_2 phase by direct high-temperature reaction calorimetry.

2.3. The Sc–Th system

The Sc–Th system is consisted of two solution phases (αTh and αSc phases) and one reciprocal solution phase (βSc and βTh). Based on thermal analysis and lattice parameter measurements, the Sc–Th system was investigated by Terekhov and Sinyakova [15], who indicated that the minimum melting point occurs at 1405 °C and 24 at.% Th, rather than ∼45 at.% Th [16] and the (βSc, βTh) ↔ (αSc) + (αTh) eutectoid temperature is 1150 °C, which is substantially (∼150 °C) higher than the value estimated by Badayeva and Kuznetsova [16]. Terekhov and Sinyakova [15] also reported the solubility of Th in the (αSc) phase to be 16 at.% Th and Sc in the (αTh) phase to be 61 at.% Sc at 650 °C. Badayeva and Kuznetsova [16] reported the solubility of Sc in the (αTh) phase to be 55 at.% Sc at 600 °C. The phase diagram of the Sc–Th system reviewed by Okamoto [17] is shown in Fig. 3.

In addition, Niessen et al. [18] predicted that the enthalpies of solution at infinite dilution is zero, however, no temperature dependence was given in the work of Niessen et al. [18].

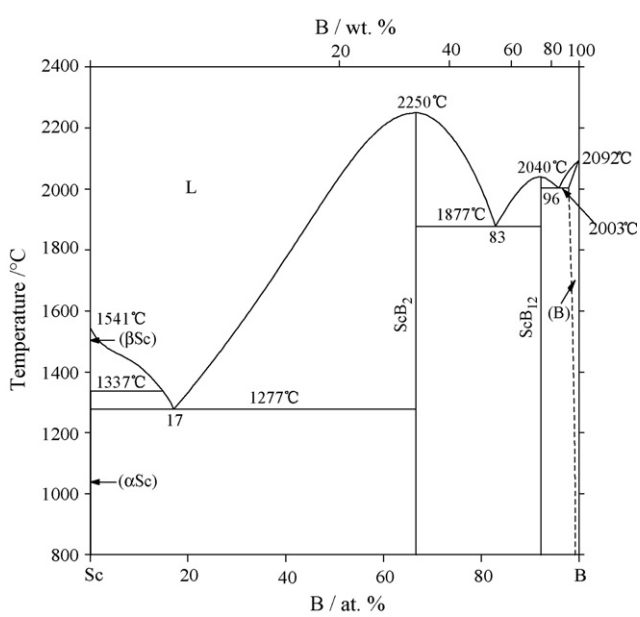

Fig. 2. The phase diagram of the Sc–B system reviewed by Spear and Liao [13].

3. Thermodynamic models

The information of stable solid phases and the used models in the Sc–M (M: Ag, B and Th) binary systems are listed in Table 1.

3.1. Solution phases

In the Ag–Sc, B–Sc and Sc–Th binary systems (A–B system), Gibbs free energies of the liquid, fcc, hcp, rhombohedral and bcc phases are described by the subregular solution model, as follows:

$$G_m^{\phi} = \sum_{i=A,B} {}^0G_i^{\phi} x_i + RT \sum_{i=A,B} x_i \ln x_i + \Delta^{\mathrm{E}} G_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [19]; x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^{\mathrm{E}} G_{\mathrm{m}}^{\phi}$ presents the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial

Table 1
The stable solid phases and the used models in the Sc–M (M: Ag [8], B [13] and Th [17]) systems

System	Phase	Strukturbericht designation	Prototype	Modeling phase	Used models
Sc–Ag	(Ag)	A1	Cu	FCC	SSM
	Ag_4Sc	D1	$MoNi_4$	$(Ag)_4(Sc)$	SM
	Ag_2Sc	C11	$MoSi_2$	$(Ag)_2(Sc)$	SM
	AgSc	B2	CsCl	(Ag)(Sc)	SM
	(αSc)	A3	Mg	HCP	SSM
	(βSc)	A2	W	BCC	SSM
Sc–B	(βB)	–	βB	Rhombohedral	SSM
	B_2Sc	C32	AlB_2	$(B)_2(Sc)$	SM
	$B_{12}Sc$	D2	$Mn_{12}Th$	$(B)_{12}(Sc)$	SM
	(αSc)	A3	Mg	HCP	SSM
	(βSc)	A2	W	BCC	SSM
Sc–Th	(αSc)	A3	Mg	HCP	SSM
	(βSc)	A2	W	BCC	SSM
	(αTh)	A1	Cu	FCC	SSM
	(βTh)	A2	W	BCC	SSM

Note: SSM, subregular solution model; SM, sublattice model.

[20] as:

$$\Delta^{E}G_m^{\phi} = x_A x_B \sum_{m=0}^{n} {}^{m}L^{\phi}(x_A - x_B)^m, \quad (2)$$

$${}^{i}L_{A,B}^{\phi} = a + bT, \quad (3)$$

where ${}^{i}L_{A,B}^{\phi}$ is the binary interaction parameter, and the coefficients of a and b are constants to be evaluated on the basis of available experimental data.

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds of the AgSc, Ag_2Sc, Ag_4Sc, ScB_2 and ScB_{12} phases in the Sc–M (M: Ag, B and Th) binary systems are stoichiometric phases. The Gibbs free energy per molar of formula unit Sc_pM_q can be expressed, as follows:

$$G_{Sc:M}^{Sc_pM_q} = p^0G_M^{SER} + q^0G_{Sc}^{SER} + \Delta^0G_f^{Sc_pM_q}, \quad (4)$$

where the $\Delta^0G_f^{Sc_pM_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is described as:

$$\Delta^0G_f^{Sc_pM_q} = a' + b'T, \quad (5)$$

where the parameters of a' and b' are evaluated in the present work.

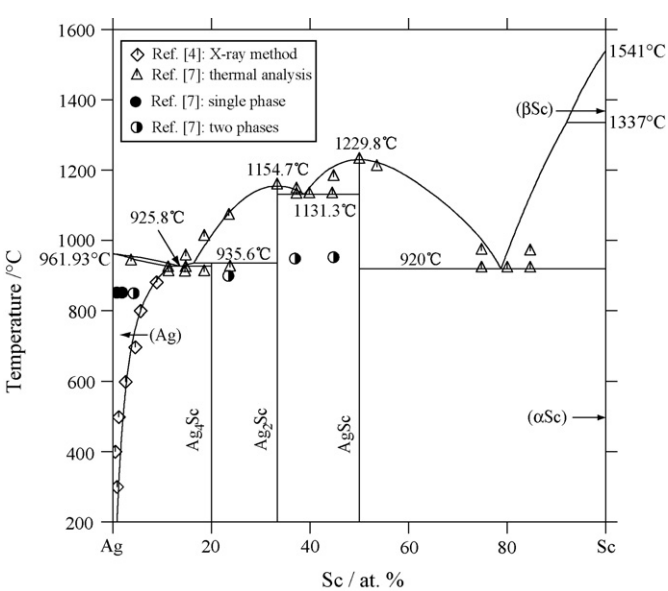

Fig. 4. The calculated phase diagram of the Sc–Ag system compared with the experimental data [4,7].

Table 2
Thermodynamic parameters in the Sc–Ag system assessed in the present work

System	Parameters in each phase (J/mol)
Sc–Ag	Liquid phase, format (Ag, Sc)
	${}^{0}L_{Ag,Sc}^{Liq} = -85110 + 23.756T$
	${}^{1}L_{Ag,Sc}^{Liq} = -23844 + 3.906T$
	${}^{2}L_{Ag,Sc}^{Liq} = -9598 + 10.154T$
	${}^{3}L_{Ag,Sc}^{Liq} = 65340 - 26.483T$
	Fcc phase, format (Ag, Sc)
	${}^{0}G_{Sc}^{Fcc} = {}^{0}G_{Sc}^{hcp} + 5000$
	${}^{0}L_{Ag,Sc}^{Fcc} = -62054 + 8.75T$
	${}^{1}L_{Ag,Sc}^{Fcc} = -77996 + 33.955T$
	${}^{2}L_{Ag,Sc}^{Fcc} = 40.351T$
	Bcc phase, format (Ag, Sc)
	${}^{0}L_{Ag,Sc}^{Bcc} = 50000$
	Hcp phase, format (Ag, Sc)
	${}^{0}L_{Ag,Sc}^{Hcp} = 50000$
	AgSc phase, format $(Ag)_{0.5}(Sc)_{0.5}$
	${}^{0}G_{Ag:Sc}^{AgSc} = 0.5 \times {}^{0}G_{Ag}^{fcc} + 0.5 \times {}^{0}G_{Sc}^{hcp} - 25594 + 3.11T$
	Ag_2Sc phase, format $(Ag)_{0.667}(Sc)_{0.333}$
	${}^{0}G_{Ag:Sc}^{Ag_2Sc} = 0.667 \times {}^{0}G_{Ag}^{fcc} + 0.333 \times {}^{0}G_{Sc}^{hcp} - 27927 + 5.558T$
	Ag_4Sc phase, format $(Ag)_{0.8}(Sc)_{0.2}$
	${}^{0}G_{Ag:Sc}^{Ag_4Sc} = 0.8 \times {}^{0}G_{Ag}^{fcc} + 0.2 \times {}^{0}G_{Sc}^{hcp} - 28203 + 12.212T$

Note: ${}^{0}G_{Sc}^{Fcc}$ is taken from the data given in Ref. [22].

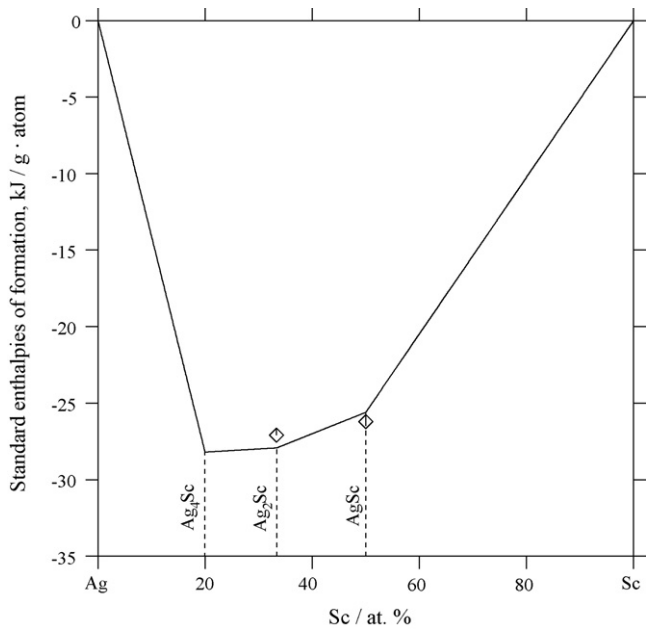

Fig. 5. The calculated standard enthalpies of formation with reference states of Ag (fcc) and αSc (hcp) in the Sc–Ag system compared with the experimental data [9].

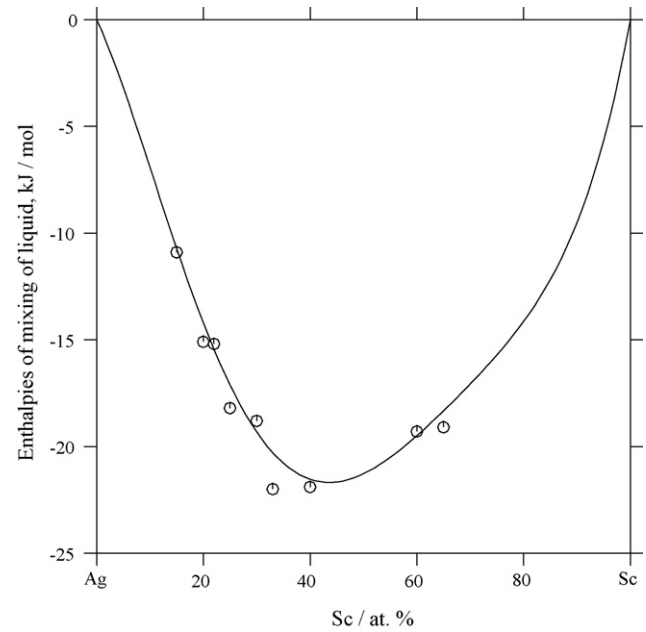

Fig. 6. The calculated enthalpies of mixing of liquid phase at 1200 °C with reference states of pure elements of Ag (liquid) and Sc (liquid) in the Sc–Ag system compared with the experimental data [9].

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [21], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

Table 3
Special points of the Sc–Ag system

Invariant reaction	Reaction type	Composition (at.%) Sc				Temperature (°C)	Reference
L ↔ βSc	Melting		100			1541	[8]
			100			1538	[7]
			100			1541	This work
βSc ↔ αSc	Polymorphic		100			1337	[8]
			100			1337	This work
L ↔ αSc + AgSc	Eutectic	80.0	100	50.0		920	[7]
		78.7	100	50.0		920	This work
L ↔ AgSc	Congruent		50.0			1230	[7]
			50.0			1229.8	This work
L ↔ AgSc + Ag_2Sc	Eutectic	40.0	50.0	33.3		1130	[7]
		38.9	50.0	33.3		1131.3	This work
L ↔ Ag_2Sc	Congruent		33.3			1155	[7]
			33.3			1154.7	This work
L + Ag_2Sc ↔ Ag_4Sc	Peritectic	–	33.3	20.0		936	[8]
		11.5	33.3	20.0		920	[7]
		16.4	33.3	20.0		935.6	This work
L ↔ Ag_4Sc + (Ag)	Eutectic	12.0	20.0	10.4		926 ± 3	[4,8]
		10.0	20.0	–		910	[7]
		13.9	20.0	11.5		925.8	This work
L ↔ (Ag)	Melting		0			961.9	[8]
			0			960.5	[7]
			0			961.9	This work

Table 4
Thermodynamic parameters in the Sc–B system assessed in the present work

System	Parameters in each phase (J/mol)
Sc–B	Liquid phase, format (B, Sc) ${}^{0}L^{\mathrm{Liq}}_{\mathrm{B,Sc}} = -210009 - 7.124T$ ${}^{1}L^{\mathrm{Liq}}_{\mathrm{B,Sc}} = -46560 - 40T$ ${}^{2}L^{\mathrm{Liq}}_{\mathrm{B,Sc}} = 24757 - 20.493T$ Bcc phase, format (B, Sc) ${}^{0}L^{\mathrm{Bcc}}_{\mathrm{B,Sc}} = 0$ Hcp phase, format (B, Sc) ${}^{0}L^{\mathrm{Hcp}}_{\mathrm{B,Sc}} = 0$ B_2Sc phase, format $(B)_{0.667}(Sc)_{0.333}$ ${}^{0}G^{\mathrm{B_2Sc}}_{\mathrm{B:Sc}} = 0.667 \times {}^{0}G^{\beta}_{\mathrm{B}} + 0.333 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} - 102330 + 8.533T$ $B_{12}Sc$ phase, format $(B)_{0.9231}(Sc)_{0.0769}$ ${}^{0}G^{\mathrm{B_{12}Sc}}_{\mathrm{B:Sc}} = 0.9231 \times {}^{0}G^{\beta}_{\mathrm{B}} + 0.0769 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} - 40974 + 4.453T$ Rhombohedral phase, format (B, Sc) ${}^{0}G^{\beta\mathrm{B}}_{\mathrm{Sc}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} + 5000$ ${}^{0}G^{\beta\mathrm{B}}_{\mathrm{B,Sc}} = -207332 + 25.492T$ ${}^{1}G^{\beta\mathrm{B}}_{\mathrm{B,Sc}} = -195201$

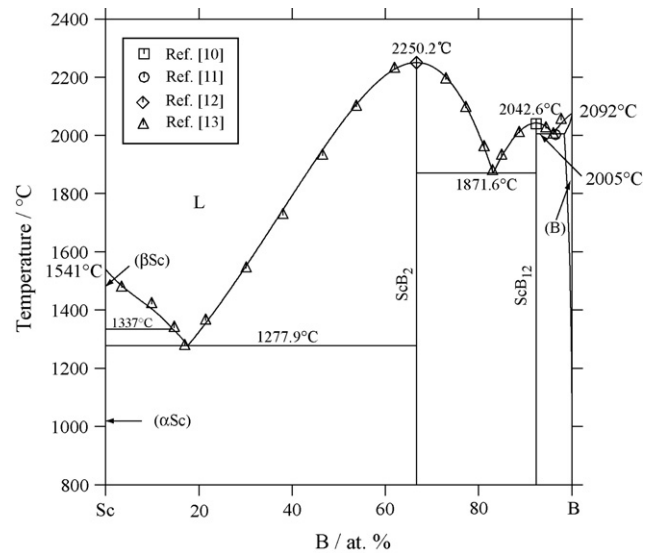

Fig. 7. The calculated phase diagram of the Sc–B system compared with the experimental data [10–13].

4.1. The Sc–Ag system

The calculated Sc–Ag phase diagram compared with the experimental data is shown in Fig. 4. It is seen that the calculated results are in reasonable agreement with the experimental data [4,7]. However, the calculated composition of the liquid phase in the peritectic reaction ($L + Ag_2Sc \leftrightarrow Ag_4Sc$) is 16.3 at.% Sc, which is slightly higher than the value of 11.5 at.% Sc proposed by Stapf et al. [7], the calculated temperature of the peritectic reaction basiclly agrees with the peritectic temperature proposed by Gschneidner and Calderwood [8], thus the calculated results are considered to be acceptable. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sc–Ag system are given in Table 2, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 3.

The calculated standard enthalpies of formation with reference states of the Ag (fcc) and αSc (hcp) phases compared with the experimental data are presented in Fig. 5, and a comparison of enthalpies of mixing of liquid at 1200 °C with reference states of pure elements of Ag (liquid) and Sc (liquid) between the calculated and experimental results is shown in Fig. 6. It is seen that a good agreement on the thermodynamic properties is obtained between the calculated results and the experimental data.

Table 5
Special points of the Sc–B system

Invariant reaction	Reaction type	Composition (at.%) B			Temperature (°C)	Reference
$L \leftrightarrow \beta B$	Melting		100		2092	[13]
			100		2092	This work
$L \leftrightarrow \beta B + ScB_{12}$	Eutectic	96.0	96.5	92.3	2003 ± 50	[11,13]
		95.9	98.4	92.3	2005.9	This work
$L \leftrightarrow ScB_{12}$	Congruent		92.3		2040 ± 50	[10]
			92.3		2042.6	This work
$L \leftrightarrow ScB_2 + ScB_{12}$	Eutectic	83.0	66.7	92.3	1877 ± 75	[13]
		83.1	66.7	92.3	1871.6	This work
$L \leftrightarrow ScB_2$	Congruent		66.7		2250 ± 50	[12]
			66.7		2250.2	This work
$L \leftrightarrow ScB_2 + \alpha Sc$	Eutectic	17.0	66.7	0	1277 ± 100	[13]
		17.6	66.7	0	1277.9	This work
$\beta Sc \leftrightarrow \alpha Sc$	Polymorphic		0		1337	[13]
			0		1337	This work
$L \leftrightarrow \beta Sc$	Melting		0		1541	[13]
			0		1541	This work

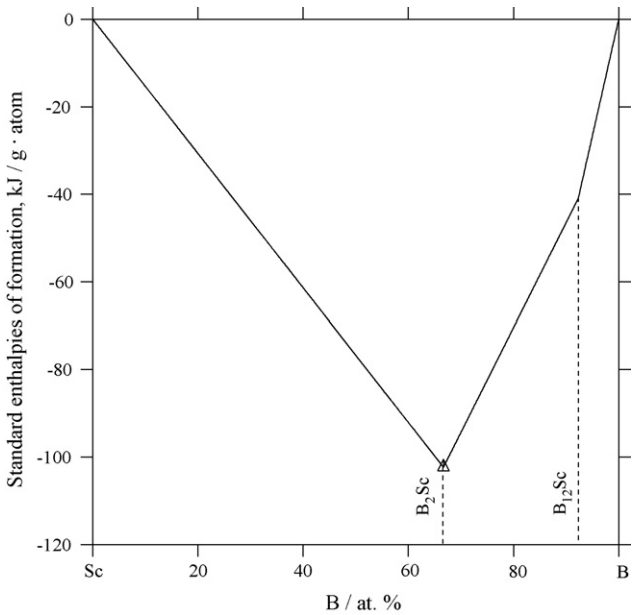

Fig. 8. The calculated standard enthalpies of formation with reference states of βB (rhombohedral) and αSc (hcp) in the Sc–B system compared with experimental data [14].

4.2. The Sc–B system

The calculated Sc–B phase diagram compared with the experimental data is shown in Fig. 7. It is seen that the calculated results agree with the experimental data of the phase equilibria from the literatures [10–13]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sc–B system are given in Table 4, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 5.

The calculated standard enthalpies of formation with reference states of the βB (rhombohedral) and αSc (hcp) phases compared with the experimental data are presented in Fig. 8. It is seen that an excellent agreement on the thermodynamic properties is obtained between the calculated results and the experimental data.

Table 6
Thermodynamic parameters in the Sc–Th system assessed in the present work

System	Parameters in each phase (J/mol)
Sc–Th	Liquid phase, format (Sc, Th) ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Sc,Th}} = -3711 + 9.084T$ ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Sc,Th}} = -4000 - 4.502T$ ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Sc,Th}} = -715 - 0.415T$ Bcc phase, format (Sc, Th) ${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Sc,Th}} = -10379 + 15.265T$ ${}^{1}L^{\mathrm{Bcc}}_{\mathrm{Sc,Th}} = -12304 + 4.004T$ Hcp phase, format (Sc, Th) ${}^{0}G^{\mathrm{Hcp}}_{\mathrm{Th}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Th}} + 5000$ ${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Sc,Th}} = 168 + 5.591T$ ${}^{1}L^{\mathrm{Hcp}}_{\mathrm{Sc,Th}} = -74052 + 49.069T$ ${}^{2}L^{\mathrm{Bcc}}_{\mathrm{Sc,Th}} = -6.2T$ Fcc phase, format (Sc, Th) ${}^{0}G^{\mathrm{Fcc}}_{\mathrm{Sc}} = {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} + 5000$ ${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Sc,Th}} = -60790 + 43.548T$ ${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Sc,Th}} = 27413 - 27.594T$ ${}^{2}L^{\mathrm{Fcc}}_{\mathrm{Sc,Th}} = 6905 - 11.9T$

Note: ${}^{0}G^{\mathrm{Fcc}}_{\mathrm{Sc}}$ is taken from the data given in Ref. [22].

4.3. The Sc–Th system

The calculated Sc–Th phase diagram compared with the experimental data is shown in Fig. 9. It is seen that the calculated results are basically in agreement with the experimental data of the phase equilibria from the literatures [15,16]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Sc–Th system are given in Table 6, and the calculated compositions and temperatures for the invariant reactions compared with the experimental data are list in Table 7. The calculated enthalpies of mixing of liquid phase at 1800 °C with reference states of pure elements of Sc (liquid) and Th (liquid) with different components are shown in Fig. 10. It is seen that the enthalpies of mixing of liquid phase are smaller in the whole composition range, which is closed to the value predicated by Niessen et al. [18].

Table 7
Special points of the Sc–Th system

Invariant reaction	Reaction type	Composition (at.%) Th				Temperature (°C)	Reference
L ↔ βSc	Melting		0			1541	[17]
			0			1541	This work
βSc ↔ αSc	Polymorphic		0			1337	[17]
			0			1337	This work
(βSc,	Eutectoid	34.0	–		20.0	∼1150	[15]
βTh) ↔ αTh + αSc		–	–		–	∼1000	[16]
		36.7	29.0		21.2	1150	This work
L ↔ βTh	Melting		100			1755	[17]
			100			1755	This work
βTh ↔ αTh	Polymorphic		100			1360	[17]
			100			1360	This work

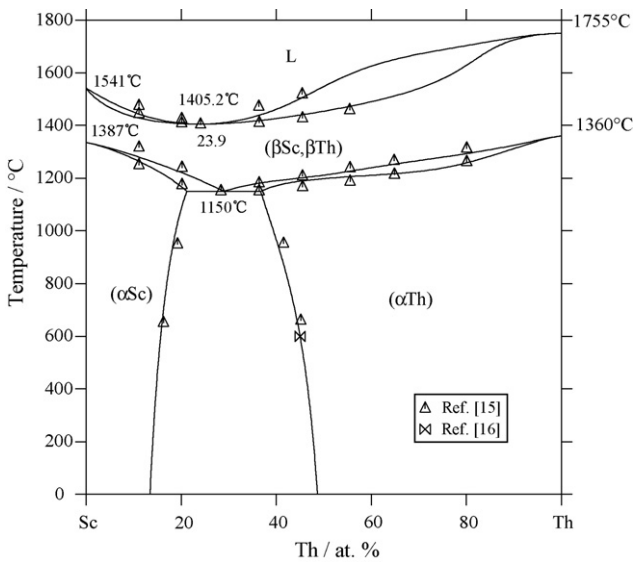

Fig. 9. The calculated phase diagram of the Sc–Th system compared with the experimental data [15,16].

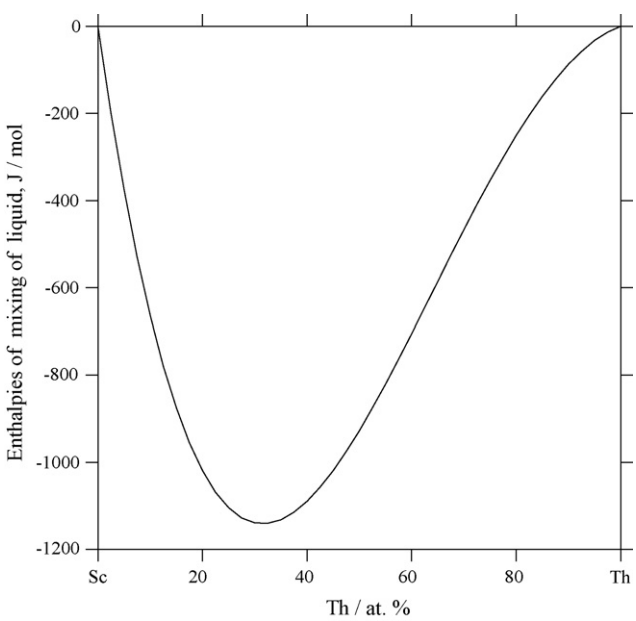

Fig. 10. The calculated enthalpies of mixing of liquid phase at 1800 °C with reference states of pure elements of Th (liquid) and Sc (liquid).

5. Conclusions

The phase diagram in the Sc–M (M: Ag, B and Th) binary systems were critically assessed by considering the experimental data including phase equilibria and thermodynamic data, and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

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